

Christian Serio

CONTACT INFORMATION	Department of Mathematics 970 Evans Hall Berkeley, CA 94720	cdserio@berkeley.edu
WEBSITE	https://cdserio.github.io	
MATHEMATICAL INTERESTS	Probability theory and mathematical/statistical physics: scaling limits, universality, Gibbs measures, line ensembles, random polymers, spin systems.	
EMPLOYMENT	University of California, Berkeley (2026–) NSF Postdoctoral Fellow	
EDUCATION	Stanford University (2021–2026) PhD in Mathematics Advisor: Amir Dembo Columbia University (2017–2021) BA in Mathematics with Honors, summa cum laude, Phi Beta Kappa	
PUBLICATIONS AND PREPRINTS	❑ “The pinned half-space Airy line ensemble,” with E. Dimitrov and Z. Yang. Submitted. ❑ “The half-space KPZ line ensemble and its scaling limit,” with S. Das. Submitted. ❑ “Scaling limit and tail bounds for a random walk model of SOS level lines,” with M. Hegde and Y. H. Kim. Submitted. ❑ “Convergence to stationary measures for the half-space log-gamma polymer,” with S. Das. <i>J. Funct. Anal.</i> 289(4): 110982 (2025). ❑ “Uniform convergence of Dyson Ferrari–Spohn diffusions to the Airy line ensemble,” with E. Dimitrov. <i>Ann. Inst. H. Poincaré Probab. Statist.</i> 61(1): 385-402 (2025). ❑ “Scaling limit for line ensembles of random walks with geometric area tilts.” <i>Electron. J. Probab.</i> 28: 1-14 (2023). ❑ “Tightness of discrete Gibbsian line ensembles.” <i>Stoch. Process. Their Appl.</i> 159: 225-285 (2023). ❑ “Tightness of Bernoulli Gibbsian line ensembles,” with E. Dimitrov, X. Fang, L. Fesser, C. Teitler, A. Wang, and W. Zhu. November 2021. <i>Electron. J. Probab.</i> 26: 1-93 (2021).	
INVITED TALKS	❑ Joint Mathematics Meetings, Special Session on Random Tilings, Random Permutations, and Particle Systems, Washington, DC, January 2026. ❑ Northeast Probability Seminar, CUNY, November 2025. ❑ Los Angeles Probability Forum, November 2025. ❑ University of Chicago, Probability Seminar, October 2025. ❑ Columbia University, Probability Seminar, October 2025. ❑ University of Utah, Stochastics Seminar, March 2025. ❑ Stanford University, Probability Seminar, October 2023. ❑ Columbia University, Columbia Probability Workshop, May 2023.	

FELLOWSHIPS AND AWARDS	❑ NSF Mathematical Sciences Postdoctoral Research Fellowship, 2026–2029. ❑ ARCS Scholar Award, Northern California Chapter of ARCS Foundation, 2024–2026. ❑ John Dash Van Buren Jr. Prize in Mathematics, Columbia College, April 2021. Awarded to one student for an outstanding senior thesis, “Tightness of discrete Gibbsian line ensembles.” ❑ Van Amringe Mathematical Prize, Columbia Mathematics Department, May 2020. Awarded to the student deemed most proficient in their class in designated mathematical subjects.
TEACHING EXPERIENCE	❑ TA at Stanford, MATH 63DM: Modern Mathematics Discrete Methods, Spring 2024, 2025 ❑ CA at Stanford, MATH 151: Intro to Probability Theory, Winter 2024 ❑ TA at Stanford, MATH 53: Ordinary Differential Equations, Winter 2023 ❑ CA at Stanford, MATH 136: Stochastic Processes, Fall 2022 ❑ CA at Stanford, MATH 158: Basic Probability and Stochastic Processes with Engineering Applications, Spring 2022 ❑ CA at Stanford, MATH 19: Calculus I, Fall 2021 ❑ TA at Columbia, MATH UN1201: Calculus III, Summer 2020 ❑ TA at Columbia, MATH GU4061–4062: Intro to Modern Analysis I & II, Spring 2019–Spring 2021
SERVICE	❑ Referee for <i>Annals of Probability</i>, <i>Forum of Mathematics: Sigma</i>, <i>Probability and Mathematical Physics</i>, and <i>Probability Theory and Related Fields</i>, 2024–present. ❑ Organizer for Stanford Student Probability Seminar, Fall 2022–Spring 2024. ❑ Columbia University Mathematics REU, Summer 2021. Graduate student mentor for Carsten Chong’s “Hurst index estimation under measurement errors” undergraduate research group.
REFERENCES	Amir Dembo, Professor, Mathematics Department, Stanford University, adembo@stanford.edu. Ivan Corwin, Professor, Mathematics Department, Columbia University, ic2354@columbia.edu. Alan Hammond, Professor, Mathematics Department, University of California, Berkeley, alanmh@stat.berkeley.edu. Evgeni Dimitrov, Assistant Professor, Mathematics Department, University of Southern California, edimitro@usc.edu. Tadashi Tokieda, Professor, Mathematics Department, Stanford University, tokieda@stanford.edu. (Teaching reference)